

transform_data

User Manual

Prepare Jira data for metrics and reports

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1 What is transform_data?

transform_data is the first module in the situation-report toolchain. It reads a raw data export from your issue tracker (Jira) and prepares the data for further analysis. The result is three **Excel files** that show how long issues spent in each workflow step and how the backlog evolved over time.

The program has a simple graphical user interface (GUI): no programming knowledge required. Select two files, click 'Run', and the Excel files are produced automatically.

What transform_data produces:

- **IssueTimes.xlsx**: All issues with milestone timestamps and minutes per workflow stage.
- **Transitions.xlsx**: Complete status history of all issues.
- **CFD.xlsx**: Daily entry counts per stage for the Cumulative Flow Diagram.

The generated Excel files are then used by the **build_reports** module to create charts and reports.

2 Prerequisites and Installation

2.1 What needs to be installed?

transform_data is delivered as a **portable package**. No separate Python installation is required.

- **Windows:** Python is already included in the package — just unzip and run.
- **macOS / Linux:** On the first launch, a Python environment is set up automatically (approx. 1 minute, internet required).

2.2 Starting the program

- **Windows:** Double-click **TransformData.bat**.
- **macOS:** Right-click **TransformData.command** → *Open* (once, to bypass Gatekeeper).
- **Linux:** In a terminal: `./TransformData.sh`

Tip (Windows): On the first launch, SmartScreen may show a warning. Click **More info** → **Run anyway**.

3 Input Files

transform_data requires exactly two files: the Jira JSON export and the workflow definition file. Both are selected in the GUI via file dialogs.

3.1 Jira JSON Export

This is an export of all issues in a Jira project in JSON format. It contains the complete status history (changelog), metadata such as issue type and creation date, and the current status for each ticket.

How to generate the export from Jira is described in the **get_data** module. The JSON file should not be edited manually.

Typical filename: ART_A.json, MYPROJECT.json, etc.

Size: Depends on the number of issues; often several MB.

3.2 Workflow Definition File

The workflow file is a plain text file (.txt) that describes how the Jira statuses in your project map to logical process steps (stages). It also defines which stage marks the start of active work (First Date) and completion (Closed Date).

Typical filename: workflow_ART_A.txt, etc.

Created by: Your technical contact or Scrum Master -- once per project, rarely changed.

4 The Workflow Definition File

The workflow file controls how transform_data interprets the Jira statuses in your project. You usually do not need to create this file yourself -- this chapter explains its structure so you can read and adjust it if needed.

4.1 Structure

Example:

```
Funnel:New:Open:To Do
Analysis:In Analysis:Estimated
Implementation:In Implementation:In Progress
Done:Canceled
<First>Analysis
<InProgress>Implementation
<Closed>Done
```

Format	Meaning
Stage:Alias1:Alias2	Stage with Jira status names mapped to it. The first name is the canonical stage name.
<First>Stage	This stage marks the start of active work (First Date).
<InProgress>Stage	This stage sets the Implementation Date.
<Closed>Stage	This stage marks completion (Closed Date).

4.2 Stage order

Stages are listed in the order they are traversed in the process. This order is used for the CFD and for computing the Closed Date when stages are skipped.

4.3 Unmapped Jira statuses

If the Jira export contains statuses not defined in the workflow file, transform_data issues a warning. Time in unmapped statuses is attributed to the last known stage (carry-forward).

5 The Graphical User Interface (GUI)

After starting, a window opens with file selection fields, options, and a log area.

5.1 Selecting files

- **JSON File** - Click 'Browse' and select your Jira export. Output folder and prefix are filled in automatically.
- **Workflow File** - Select the appropriate .txt workflow file for your project.

Tip: If you select the JSON file first, the output folder and prefix are pre-filled automatically.

5.2 Configuring output

Field	Meaning
Output Folder	Directory where the three Excel files are saved. Default: directory of the JSON file.
Prefix	Name prefix for output files. 'ART_A' produces ART_A_IssueTimes.xlsx, ART_A_Transitions.xlsx, and

5.3 Starting the transformation

Click '**Run**'. The program reads the data, computes all values, and saves the three Excel files. Progress and any warnings appear in the log area.

Warnings in the log: A warning about unmapped statuses does not mean the transformation failed -- the output files are still produced.

5.4 Language and Help

Use the **Options** menu to switch between German and English. Under **Help** → **Manual**, this manual opens in your browser.

6 Output Files

transform_data produces three Excel files used as input for build_reports.

6.1 IssueTimes.xlsx

Column	Content
Project	Project key (e.g. ART_A)
Key	Issue key (e.g. ART_A-123)
Issuetype	Issue type (e.g. Feature, Bug, Story)
Status	Current Jira status
Created Date	Issue creation date in Jira
First Date	First entry into the <First> stage
Implementation Date	First entry into the <InProgress> stage
Closed Date	Completion timestamp (empty = still open)
Stage columns	One column per stage: minutes spent in that stage
Resolution	Closure type from Jira

6.2 Transitions.xlsx

Contains the complete status history of all issues -- one entry per status change.

Column	Content
Key	Issue key
Transition	Stage name (or 'Created' for the creation timestamp)
Timestamp	Date and time of the status change

6.3 CFD.xlsx

Contains, for each calendar day, the number of issues that entered each stage on that day. build_reports accumulates these values to produce the Cumulative Flow Diagram.

7 How are Dates and Times Calculated?

7.1 Stage times

Situation	Rule
Issue created but no status change yet	Time from creation to the first status change is attributed to the initial stage.
Status change to an unmapped stage	Time continues in the last known stage (carry-forward). No time is lost.
Current status	The last known stage accumulates time until processing time.

7.2 First Date

The First Date is set when an issue first enters the stage marked **<First>**. It represents the start of active work.

Skipped First stage: If an issue enters a stage after <First> without entering it directly, that entry timestamp is used as the First Date.

7.3 Closed Date

The Closed Date is set when an issue enters the stage marked **<Closed>**. If closed and re-opened multiple times, the **last** closing timestamp is used.

Re-opened issues: If an issue is currently before <Closed>, no Closed Date is set.

Never-worked issues: Issues without a First Date receive no Closed Date and are not counted in Flow Velocity.

8 Frequently Asked Questions

Q: An issue has no First Date -- why?

A: The issue never reached the stage or any stage after it. In build_reports it is counted as 'To Do'.

Q: An issue has no Closed Date -- why?

A: Possible reasons: (1) The issue is still open. (2) It has no First Date. (3) It was re-opened after closing.

Q: The log shows a warning about unmapped statuses.

A: Some Jira statuses are not defined in the workflow file. Processing still completes.

Q: The output files are not created.

A: Check that the output folder exists and you have write permissions. Make sure the files are not open in Excel.

Q: Which file do I use for build_reports?

A: IssueTimes.xlsx is required for all metrics. CFD.xlsx is additionally needed for the Cumulative Flow Diagram.

Q: How often do I need to run transform_data?

A: Whenever you need updated data from Jira.

Q: What does 'carry-forward' mean?

A: Time in an unmapped status is attributed to the last known stage -- no time is lost.

9 Glossary

Term	Explanation
Carry-forward	Time in an unmapped status is attributed to the last known stage.
CFD	Cumulative Flow Diagram -- shows backlog evolution over time by stage.
Closed Date	The date an issue was completed.
First Date	The date active work began on an issue.
Implementation Date	The date development work started.
Issue	A ticket in the issue tracker (e.g. a Jira card).
JSON	A simple text format for structured data.
Marker	Lines in the workflow file that designate a stage as a milestone: <First>, <InProgress>, <Closed>.
Prefix	Name prefix for output files (e.g. 'ART_A').
Stage	A logical process step mapped to one or more Jira statuses.
Workflow file	Text file describing the stages and markers of a project.